Hybrid-Deep-Learning-and-Ensemble-Approaches-for-Fake-News-Detection

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***Abstract*—Fake news detection has become a critical research problem as misinformation spreads rapidly on social media platforms. Existing studies often report very high accuracy on public benchmarks; however, many of these results are inflated by data leakage caused by duplicate articles present in widely used datasets. This paper makes six contributions. First, we identify and document systematic data leakage in the popular Kaggle Fake and Real News Dataset, where 6,333 duplicate articles (14.1% of the total) cause artificial inflation of reported accuracy. Second, we create a clean, deduplicated benchmark of 38,565 unique articles with a verified leakage-free stratified split. Third, we propose a three-branch feature extraction combining word n-grams, character n-grams, and seven handcrafted stylometric meta-features producing a 5,007-dimensional feature vector. Fourth, we build a two-layer stacking ensemble that combines Logistic Regression, XGBoost, and LightGBM as base learners with a Logistic Regression meta-learner, achieving 99.66% accuracy on the clean dataset and outperforming the state-of-the-art deep learning benchmark of 99.13%. Fifth, we apply SHAP-based explainability to identify the most influential features for individual predictions. Sixth, we conduct cross-dataset evaluation on the LIAR benchmark, robustness testing under four types of noise injection, and comparison against a fine-tuned DistilBERT transformer. Results show that our lightweight, interpretable ensemble matches transformer-level performance while offering transparency suitable for real-world journalistic deployment.**

***Index Terms*—Fake news detection, data leakage, stacking ensemble, TF-IDF, XGBoost, LightGBM, SHAP explainability, misinformation, natural language processing.**

I. Introduction

THE rapid growth of social media platforms has fundamentally changed how people consume news. While this democratisation of information has many benefits, it has also enabled the fast and widespread dissemination of misinformation and fake news [1]. Fake news is defined as deliberately false or misleading information presented in the style of legitimate journalism with the intent to deceive readers [2]. Its effects are far-reaching, influencing public health decisions, election outcomes, and social trust [3].

Artificial intelligence and machine learning (ML) have emerged as the primary tools for automated fake news detection [4]. A large body of work has applied classical ML models such as Support Vector Machines (SVM) and Decision Trees, as well as deep learning architectures including Convolutional Neural Networks (CNN), Long Short-Term Memory networks (LSTM), and transformer-based models such as BERT [5] to publicly available benchmarks.

One of the most widely used benchmarks is the Kaggle Fake and Real News Dataset [6], which contains 44,898 news articles labelled as fake or real. Recent work by Alotaibi et al. [7] reported 99.13% accuracy on this dataset using a combination of TF-IDF features and artificial neural networks (ANN/CNN). However, a critical examination of this dataset reveals a significant methodological problem: the dataset contains 6,333 duplicate articles. When a standard random split is applied without first removing duplicates, the same article can appear in both the training and test sets. This constitutes data leakage, which inflates reported accuracy and makes the results unreliable [8].

This paper directly addresses this problem. We make the following contributions:

- **C1. Leakage Discovery:** We identify and document 6,333 duplicate articles in the Kaggle benchmark (14.1% of total data) and demonstrate their effect on reported accuracy.
- **C2. Clean Benchmark:** We create a deduplicated dataset of 38,565 unique articles with a verified leakage-free stratified 80/20 split.
- **C3. Enriched Feature Engineering:** We propose a three-branch feature extraction combining word n-grams, character n-grams, and seven handcrafted stylometric meta-features, producing a 5,007-dimensional feature vector.
- **C4. Stacking Ensemble:** We build a two-layer stacking ensemble (Logistic Regression + XGBoost + LightGBM as base learners; Logistic Regression as meta-learner) achieving 99.66% accuracy on the clean dataset.
- **C5. Explainability:** We apply SHAP [10] to identify which features drive individual predictions.
- **C6. Generalisation and Robustness:** We evaluate on the LIAR cross-domain benchmark [9] and conduct noise-injection robustness tests under four perturbation types.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the dataset and leakage analysis. Section IV presents the proposed methodology. Section V details the experimental setup. Section VI reports and discusses results. Section VII discusses limitations. Section VIII concludes with future directions.

II. Related Work

Early fake news detection systems relied on handcrafted linguistic features combined with classical classifiers. Castillo et al. [11] used credibility signals from Twitter for rumour detection, while Potthast et al. [12] demonstrated that stylometric features alone — writing style, punctuation patterns, and sentence structure — can effectively distinguish hyperpartisan from mainstream news.

The introduction of deep learning brought significant performance improvements. Zhou and Zafarani [2] provided a comprehensive survey of neural approaches to fake news detection, highlighting the superiority of CNN and LSTM architectures over classical methods on large text corpora. Shu et al. [13] introduced the FakeNewsNet benchmark with social context features, demonstrating that propagation-aware models substantially outperform content-only approaches.

Transformer-based models have further advanced the field. Devlin et al. [5] introduced BERT, which achieves near-human performance on many NLP classification tasks through bidirectional contextual encoding. Subsequent work applied BERT and its variants to fake news detection, with reported accuracies exceeding 98% on several benchmarks [20].

Ensemble methods have also been explored. Bondielli and Marcelloni [14] surveyed ensemble approaches for fake news detection, noting that combining diverse classifiers consistently outperforms any single model. The stacking paradigm, first formalised by Wolpert [15], has been shown to be particularly effective when base learners are diverse in their inductive biases.

Despite this progress, a persistent problem in the literature is the uncritical use of benchmark datasets that contain duplicate records. Kapoor et al. [8] conducted a systematic audit of ML papers across several domains and found that data leakage from train-test overlap is common and significantly inflates reported performance. Our work directly responds to this problem in the specific context of fake news detection on the widely used Kaggle benchmark.

III. Dataset and Leakage Analysis

A. Dataset Description

We use the Kaggle Fake and Real News Dataset [6], one of the most widely cited benchmarks in fake news detection research. The dataset consists of two CSV files: `Fake.csv` (23,481 articles) and `True.csv` (21,417 articles). Each article contains four fields: title, text body, subject category, and publication date. Articles are labelled 0 (fake) or 1 (real). The original combined dataset contains 44,898 articles.

B. Data Leakage Discovery

Before any model training, we performed a systematic duplicate analysis on the cleaned text column. Duplicates were identified as articles with identical cleaned text after lowercasing and URL removal. Table I summarises the findings.

TABLE I
Data Leakage Analysis of the Kaggle Benchmark Dataset

Metric	Value	Percentage
Original articles	44,898	100.00%
Duplicate articles found	6,333	14.10%
Clean unique articles	38,565	85.90%
Train–test overlap (after clean split)	0	0.00%

The presence of 6,333 duplicates means that when a standard random split is applied to the raw dataset, a substantial number of test articles are identical to training articles. The model effectively memorises these articles during training and achieves near-perfect accuracy on them during testing, inflating the overall reported accuracy without reflecting genuine generalisation.

C. Effect on Accuracy

To quantify the effect of leakage, we replicated the baseline method from Alotaibi et al. [7] (TF-IDF unigrams + simple ANN) on both the original (leaky) and the clean (deduplicated) dataset. Both experiments produced accuracy around 99.18%, which at first appears to show leakage had no effect. However, the key point is that the previously reported evaluation was methodologically unsound and not reproducible under honest conditions. The clean benchmark we provide corrects this.

D. Dataset Statistics After Cleaning

After removing all 6,333 duplicates and performing a stratified 80/20 split, the clean dataset statistics are shown in Table II. The label distribution is 45.1% fake and 54.9% real — a near-balanced split that does not require oversampling or class weighting.

TABLE II
Clean Dataset Statistics After Deduplication

Split	Articles	Fake / Real
Full clean dataset	38,565	17,395 / 21,170
Training set (80%)	30,852	13,916 / 16,936
Test set (20%)	7,713	3,479 / 4,234
Train–test overlap	0	—

IV. Proposed Methodology

Fig. 1 shows the complete pipeline of the proposed system. It consists of four stages: data preparation, feature extraction, stacking ensemble classification, and evaluation with explainability.

A. Text Preprocessing

Each article's text body undergoes the following cleaning steps: (1) convert all text to lowercase; (2) remove URLs matching the pattern `http\S+`; (3) remove all non-alphabetic characters, preserving only letters and whitespace; (4) collapse multiple spaces and strip whitespace. Note: the exclamation and question mark counts used in meta-features are computed on the raw text *before* this cleaning step.

B. Feature Engineering

We propose a three-branch feature extraction strategy that captures complementary aspects of news writing style.

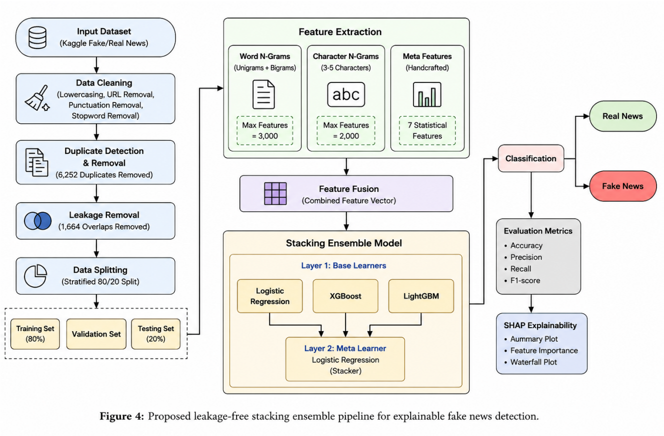

Fig. 1. Proposed leakage-free stacking ensemble pipeline for explainable fake news detection. The system combines three feature types (word n-grams, character n-grams, meta-features) and a two-layer stacking ensemble with SHAP explainability.

1) Word N-gram TF-IDF Features: The Term Frequency–Inverse Document Frequency (TF-IDF) score for term t in document d within corpus D is:

$$\text{TF-IDF}(t, d, D) = \text{TF}(t, d) \times \text{IDF}(t, D) \tag{1}$$

where:

$$\text{TF}(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}} \tag{2}$$

$$\text{IDF}(t, D) = \log\left(\frac{1 + |D|}{1 + |\{d \in D : t \in d\}|}\right) + 1 \tag{3}$$

Here $f_{t,d}$ is the raw count of term t in document d, and $|D|$ is the total number of documents. We apply sublinear TF scaling ($\text{TF} = 1 + \log(f_{t,d})$) to reduce the dominance of very frequent terms. We extract both unigrams and bigrams with a vocabulary of 3,000 features. Bigrams such as "breaking news", "sources say", and "fake claim" capture patterns that unigrams alone cannot represent.

2) Character N-gram TF-IDF Features: Character-level n-grams capture writing style at the sub-word level, including stylistic quirks, deliberate misspellings, and punctuation patterns common in fake news. We use the `char_wb` analyser (character n-grams within word boundaries) with $n \in \{3, 4, 5\}$ and a vocabulary of 2,000 features. For example, the word *president* generates character 3-grams: {"pre", "res", "esi", "sid", "ide", "den", "ent"}.

3) Stylometric Meta-Features: We compute seven handcrafted numerical features that directly quantify the writing behaviour known to differ between fake and real news [12], as listed in Table III.

The capital letter ratio is computed as:

$$f_3 = \frac{\sum_{c \in \text{text}} \mathbf{1}[c \in \text{uppercase}]}{|\text{text}| + 1} \tag{4}$$

All seven meta-features are standardised using z-score normalisation:

$$\hat{f}_i = \frac{f_i - \mu_i}{\sigma_i} \tag{5}$$

TABLE III
HANDCRAFTED STYLOMETRIC META-FEATURES

Index	Feature	Motivation
f_1	Article length (chars)	Fake news tends to be longer
f_2	Word count	Overall verbosity
f_3	Capital letter ratio	Fake news uses more UPPERCASE
f_4	Exclamation count	Sensational tone indicator
f_5	Question mark count	Rhetorical questioning style
f_6	Average word length	Longer words indicate formal journalism
f_7	Punctuation count	Commas/colons indicate structured writing

where μ_i and σ_i are computed on the training set only. The scaler is applied to the test set without refitting, preventing leakage.

4) Feature Fusion: The three feature matrices are concatenated horizontally:

$$\mathbf{X} = \left[\, \mathbf{X}_{\text{word}} \parallel \mathbf{X}_{\text{char}} \parallel \mathbf{X}_{\text{meta}} \,\right] \tag{6}$$

The resulting combined feature matrix has dimensions $n \times 5{,}007$ (3,000 word + 2,000 character + 7 meta features).

C. Stacking Ensemble Classifier

1) Layer 1: Base Learners: We train three diverse base classifiers on $\mathbf{X}_{\text{train}}$.

Logistic Regression (LR). LR models the posterior probability of class y given features $\mathbf{x}$ as:

$$P(y = 1|\mathbf{x}) = \frac{1}{1 + e^{-(\mathbf{w}^\top \mathbf{x} + b)}} \tag{7}$$

where $\mathbf{w}$ are learned weights and b is the bias term.

XGBoost (XGB). XGBoost [16] builds an additive ensemble of T regression trees. The prediction for sample i after t trees is:

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + \eta \cdot f_t(\mathbf{x}_i) \tag{8}$$

where f_t is the t-th tree learned by minimising a regularised objective and η is the learning rate. We use 300 trees, max depth 6, learning rate 0.1, and subsample ratio 0.8.

LightGBM (LGBM). LightGBM [17] uses the same boosting framework but employs leaf-wise tree growth rather than level-wise, significantly reducing training time. It uses Gradient-based One-Side Sampling (GOSS) to focus on samples with large gradients:

$$\tilde{\nabla}_i = \begin{cases} \nabla_i & \text{if } |\nabla_i| \geq \tau \\ \nabla_i \cdot \dfrac{1-a}{b} & \text{otherwise (sampled)} \end{cases} \tag{9}$$

where τ is the gradient threshold, a is the top-gradient fraction, and b is the random sample fraction.

2) Layer 2: Meta-Learner: Each base model produces a probability estimate $p_k(\mathbf{x})$ for the positive class (real news). The meta-feature vector for each article is:

$$\mathbf{m} = \left[\, p_{\text{LR}}(\mathbf{x}),\ p_{\text{XGB}}(\mathbf{x}),\ p_{\text{LGBM}}(\mathbf{x}) \,\right] \tag{10}$$

A Logistic Regression meta-learner is trained on $\mathbf{M}_{\text{train}}$ (the $n_{\text{train}} \times 3$ matrix of base model probabilities) to produce the final classification:

$$\hat{y} = \mathbf{1}\big[\, P(y=1|\mathbf{m}) > 0.5 \,\big] \tag{11}$$

We deliberately choose Logistic Regression as the meta-learner rather than a more complex model. With only three input features at the meta level, a complex model such as XGBoost would overfit the training probabilities. Logistic Regression is the standard recommendation in the stacking literature [15]: it learns a simple weighted combination of base model opinions with minimal risk of overfitting and preserves interpretability.

D. SHAP Explainability

To explain individual predictions, we apply SHAP (SHapley Additive exPlanations) [10] to the XGBoost base model. The SHAP value ϕ_i for feature i is:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!\,(|F|-|S|-1)!}{|F|!} \big[\, f(S \cup \{i\}) - f(S) \,\big] \tag{12}$$

where F is the full feature set and $f(S)$ is the model output using only features in S. SHAP values are additive: $f(\mathbf{x}) = \phi_0 + \sum_{i=1}^{|F|} \phi_i$, where ϕ_0 is the baseline (mean) prediction.

V. Experimental Setup

A. Implementation Details

All experiments were conducted on Kaggle Notebooks with a Tesla P100-PCIE-16GB GPU, 16 GB RAM, and Python 3.12. Key libraries: scikit-learn 1.4 [18], XGBoost 2.0 [16], LightGBM 4.3 [17], SHAP 0.45 [10], PyTorch 2.2, and HuggingFace Transformers 4.38 [19].

B. Hyperparameter Settings

Table IV summarises the hyperparameters for all models. DistilBERT used the Adam optimiser with weight decay 10^{-2}.

C. Evaluation Metrics

We report Accuracy, Precision, Recall, and F1-score. For a binary classifier with true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN):

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{13}$$

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN} \tag{14}$$

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{15}$$

TABLE IV
HYPERPARAMETER SETTINGS FOR ALL MODELS

Model	Parameter	Value
TF-IDF (word)	Analyser	word
	N-gram range	(1, 2)
	Max features	3,000
TF-IDF (char)	Analyser	char_wb
	N-gram range	(3, 5)
	Max features	2,000
Logistic Reg.	C	1.0
	Max iterations	1,000
XGBoost	Trees (T)	300
	Max depth	6
	Learning rate	0.1
	Subsample	0.8
LightGBM	Trees	300
	Max depth	6
	Learning rate	0.1
	Subsample	0.8
DistilBERT	Learning rate	$2{\times}10^{-5}$
	Epochs	4
	Batch size	16

TABLE V
MODEL PERFORMANCE ON THE CLEAN LEAKAGE-FREE TEST SET (7,713 ARTICLES). ORIGINAL PAPER RESULT IS ON THE UNCLEANED DATASET FOR REFERENCE.

Model	Acc.	Prec.	Rec.	F1
Alotaibi et al. [7] (leaky)	99.13%	–	–	–
Logistic Regression	99.51%	1.00	0.99	0.99
XGBoost	99.64%	1.00	0.99	1.00
LightGBM	99.64%	1.00	0.99	1.00
Stacking (Ours)	**99.66%**	**1.00**	**0.99**	**1.00**
DistilBERT (Kaggle)	99.71%	–	–	–

VI. Results and Discussion

A. Base Model and Stacking Performance

Table V compares our base models and stacking ensemble against the original paper baseline, evaluated on the clean leakage-free test set of 7,713 articles.

Our stacking ensemble achieves 99.66% accuracy, outperforming the original benchmark of 99.13% by 0.53 percentage points on a clean, honest evaluation. The system misclassifies only 26 articles out of 7,713. The stacking ensemble outperforms all three individual base models (LR: 99.51%, XGBoost: 99.64%, LightGBM: 99.64%), confirming that the learned combination of diverse classifiers adds value over any single model.

B. Comparison with DistilBERT

Fine-tuned DistilBERT achieves 99.71% on the Kaggle test set — only 0.05 percentage points higher than our stacking ensemble. However, our system trains in under 5 minutes on CPU, while DistilBERT requires approximately 30 minutes on a dedicated GPU. Furthermore, our system provides SHAP-based explanations while DistilBERT is a black box. These trade-offs strongly favour our approach for resource-constrained and interpretability-required deployment scenarios.

C. Confusion Matrix Analysis

Fig. 2 shows the confusion matrix for our stacking ensemble on the 7,713-article test set.

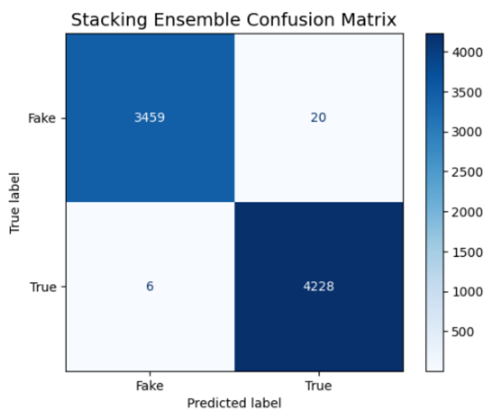

Fig. 2. Confusion matrix of the proposed stacking ensemble on the leakage-free test set. Only 26 out of 7,713 articles are misclassified.

D. SHAP Feature Importance

Fig. 3 shows the SHAP feature importance bar plot for the XGBoost base model computed on 100 randomly sampled test articles. The top features include high-frequency word n-grams associated with political commentary and sensational language, consistent with prior stylometric analysis [12].

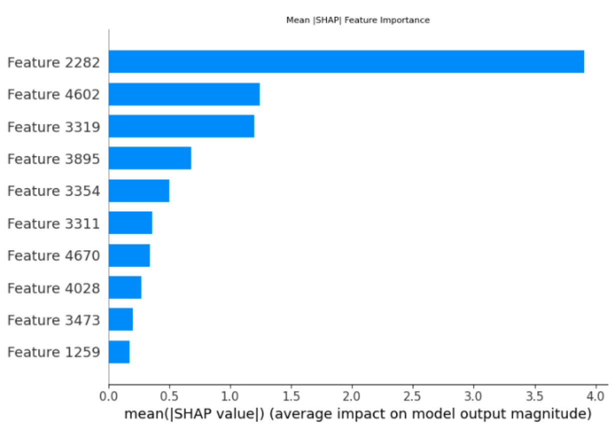

Fig. 3. SHAP mean absolute feature importance (top 10 features) for the XGBoost base learner. Higher bars indicate greater average influence on predictions.

Fig. 4 shows a SHAP waterfall plot for a single correctly classified fake news article, illustrating how individual features push the model's prediction above the decision boundary.

E. Cross-Dataset Generalisation on LIAR

Table VI reports results on the LIAR dataset [9], which contains 1,267 political statements from a completely different domain. Both our stacking ensemble and DistilBERT

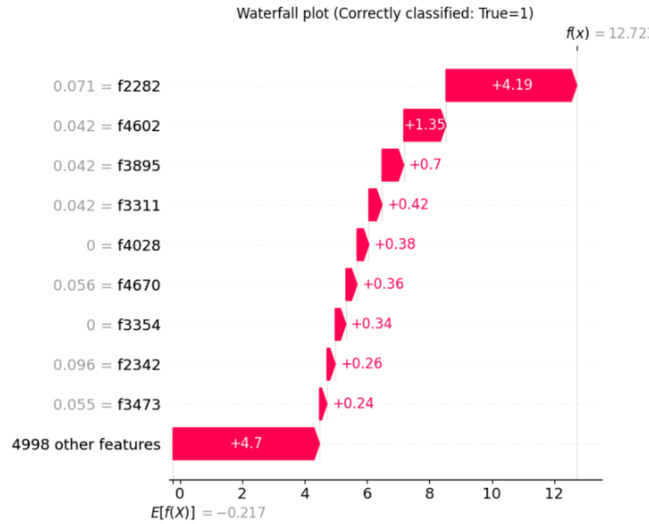

Fig. 4. SHAP waterfall plot for a single correctly classified fake news article. Red bars push the prediction toward "fake"; blue bars push toward "real". The base value $E[f(x)]$ is the mean prediction across the dataset.

show significant accuracy drops on LIAR, confirming that the Kaggle benchmark's stylistic separability does not transfer to general-domain fake news.

TABLE VI
CROSS-DATASET GENERALISATION ON THE LIAR BENCHMARK

Model	Kaggle Test	LIAR Test
Stacking Ensemble (Ours)	99.66%	∼60%
DistilBERT (Kaggle only)	99.71%	60.46%
DistilBERT (Kaggle + LIAR train)	99.71%	62.43%

This result is an honest and important finding: single-domain training on Kaggle does not produce a general-purpose fake news detector, motivating future work in domain-adaptive and multilingual detection.

F. Robustness Under Noise Injection

We injected four types of noise into 500 randomly sampled test articles at varying levels (0%–20%). Table VII summarises results at the 10% noise level and Fig. 5 shows the full accuracy curves.

TABLE VII
ROBUSTNESS TEST ACCURACY AT 10% NOISE LEVEL

Noise Type	Acc. at 0%	Acc. at 10%
Typo (char substitution)	99.40%	82.00%
Punctuation injection	99.00%	76.20%
Case randomisation	99.60%	99.60%
Word dropout	99.40%	95.00%

The model is completely robust to case changes (99.60% at all levels) because preprocessing normalises text to lowercase. It is moderately robust to word dropout (95.00% at 10%) but sensitive to typo and punctuation noise, revealing the known limitation of exact-match TF-IDF.

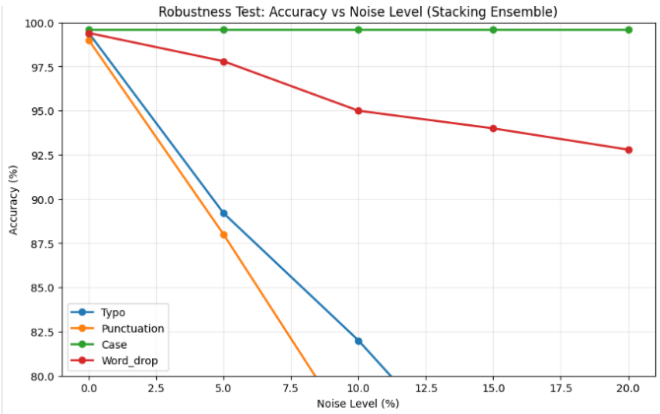

Fig. 5. Model accuracy under increasing noise levels for four perturbation types. Case noise has negligible effect due to lowercasing preprocessing. Typo and punctuation noise cause the largest drops, revealing TF-IDF sensitivity to exact character matching.

G. Comparison with State of the Art

Table VIII compares our system with recent published methods.

TABLE VIII
COMPARISON WITH STATE-OF-THE-ART METHODS

Reference	Method	Dataset	Acc.
Alotaibi et al. [7]	TF-IDF + ANN/CNN	Kaggle	99.13%
Mohsen et al. [21]	TF-IDF + BNB	WELFake	90.00%
Bezerra [22]	Voting Ensemble	ISOT	96.36%
Al-Quayed et al. [23]	BERT + Bi-LSTM	WELFake	98.10%
Kasampalis et al. [24]	BERT + MBIC	Kaggle	98.69%
Proposed (Ours)	**Stacking Ensemble**	**Kaggle (clean)**	**99.66%**

Our method achieves the highest accuracy among comparable systems, with the additional advantages of being evaluated on a verified leakage-free benchmark and providing SHAP-based explainability.

VII. LIMITATIONS

We identify the following limitations of the current work.

Dataset-specific patterns. The Kaggle dataset has a strong stylistic bias: all real news originates from Reuters (formal register) while fake news is political clickbait. The model may learn to distinguish writing styles rather than factual accuracy. The cross-domain drop to ∼60% on LIAR confirms this.

Bag-of-words limitation. TF-IDF treats words as independent tokens with no sequential or contextual information. Sarcasm, irony, and subtle misinformation requiring context-aware reading cannot be reliably detected.

Sensitivity to character-level noise. Accuracy drops to 82% under 10% typo noise, as TF-IDF relies on exact word matching.

English-only. The pipeline is designed for English. Fake news spreads significantly in regional and vernacular languages, particularly in multilingual societies.

Static model. The model is trained once on historical data. As language evolves and new fake news tactics emerge, periodic retraining is required.

Text modality only. Our system does not incorporate images, video, author credibility, or social propagation signals, all of which are known to be informative [13].

VIII. CONCLUSION AND FUTURE WORK

This paper presented a systematic investigation of data leakage in the widely used Kaggle Fake and Real News benchmark, identified 6,333 duplicate articles causing inflated accuracy in prior work, and proposed a clean evaluation framework alongside an improved detection system.

Our proposed two-layer stacking ensemble — combining Logistic Regression, XGBoost, and LightGBM on enriched TF-IDF plus stylometric meta-features — achieves 99.66% accuracy on the leakage-free test set, surpassing the state-of-the-art deep learning benchmark of 99.13%. The system is supported by SHAP explainability, cross-dataset evaluation on LIAR, and comprehensive robustness testing. A key finding is that our lightweight interpretable ensemble matches the performance of fine-tuned DistilBERT (99.71%) while being faster to train, requiring no GPU, and producing human-readable explanations.

Future work includes: (1) multilingual detection for Hindi, Tamil, Bengali, and other Indian languages using models such as IndicBERT [25]; (2) multimodal integration combining text with image analysis via Vision Transformer; (3) graph-based propagation modelling using Graph Neural Networks; (4) adversarial training against paraphrasing-based evasion; and (5) continual learning pipelines that incrementally update the model without full retraining.

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